

Alex Henin

+1 (805) 703-8436 • alexhenin.com • alexhenin110@gmail.com • Long Beach, CA

SUMMARY

Aerospace Engineering student and hands-on engineer with direct experience building, assembling, and testing liquid propellant rocket systems. Strong interest in metal additive manufacturing, automated production systems, and applying machining and test experience to building ambitious hardware.

EDUCATION

Cal State Long Beach (CSULB) | Aerospace Engineering (GPA: 3.7)

Graduating 2028

Providence High School | Valedictorian (GPA: 3.85)

RELEVANT SKILLS

Software: SolidWorks, Fusion 360, Onshape, Blender, GitHub, VS Code, Excel

Programming: Python, MATLAB, Java, JavaScript, HTML, CSS

Engineering: CAD Modeling/Drafting, GD&T, DAQ, Procedure/Technical Writing, First Principles Design, Prototyping, Pipe Bending/Cutting, Sheet Metal Manufacturing, 3D Printing/Slicing, Soldering, Power Tools

Operations & Testing: Pressure & Cryogenic Operations, Procedure Writing, Hazardous Material Handling, Hot/Static Fire Testing, Cold Flow Testing, Hydrostatic Testing, Pyrotechnics Testing, Hardware Inspection, In-field Troubleshooting

EXPERIENCE

Beach Launch Team – Fluids and Propulsion Engineer – Long Beach, CA

Aug. 2025 – Present

- Designed fluid systems for LOX and kerosene liquid-fueled rocket using Darcy-Weisbach equations.
- Validated system integration and combustion stability by executing a successful hot-fire of the LR-101 engine under flight conditions and producing 800 pounds of thrust; used findings to fix problem with in-line regulators
- Performed waterflow testing across the full propellant system, collecting data to calculate discharge coefficients, validate Darcy-Weisbach pressure drops, and identify flow bottlenecks.
- Authored Inline Regulator Functionality Test Procedures, detailing configurations, hazards, and verification steps.

IPT (Integrated Procurement Technologies) – QA Specialist – Goleta, CA

June 2024 – Aug. 2024

- Inspected aerospace and defense components for FMS, DCS, USG, and OEM clients, ensuring 100% documentation compliance and preventing part rejections.
- Led a complete overhaul of the backend documentation system, redesigning part order workflows to streamline required information, reducing turnaround time and improving contract clarity.

Riviera Robotics – CNC/General Technician – Goleta, CA

Oct. 2021 – Dec. 2021

- Machined precision components for F.I.R.S.T. competition robots using CNC machinery.
- Diagnosed mechanical failures and repaired parts to restore robot performance under competition time pressure.

Kids Helping Kids – Chief Financial Officer – Santa Barbara, CA

Sept. 2023 – Feb. 2025

- Managed revenue streams, budgets, and backend (payments, website, etc) for 100% student-run nonprofit
- Led complete sell out of +2,000 seat theater and raised +\$80,000 in net profit that went directly towards fighting human trafficking and supporting vulnerable children globally ([Article](#))
- Founded and led team behind “Call to Give”, raising \$20,000 using NFC bracelets

PROJECTS

Rocket Ignitor

Jan. 2026

- Developed a reusable e-match mandrel, replacing a single-use disposable system by designing a three-pronged mechanical interface that precisely centers the e-match within the combustion chamber

NFC Bracelets

Jan. 2024

- Fabricated 300 NFC-enabled bracelets and built a custom JavaScript website to share stories of rescued children, raising \$20,000 for Kids Helping Kids.

Disaster Relief Infrastructure Project

June 2023

- Rebuilt university and provided disaster relief on Guam and nearby islands from worst hurricane in 20 years
- Manufactured four 20-ft steel light poles on-site following the worst hurricane in Guam in 20 years; performed structural analysis to validate wind load stability prior to installation

Portfolio Website

June 2023

- Coded a Single Page Application (SPA) in JavaScript, eliminating load times across a multi-section portfolio
- Implemented custom features including image carousels and a pulsating header; used GitHub for version control